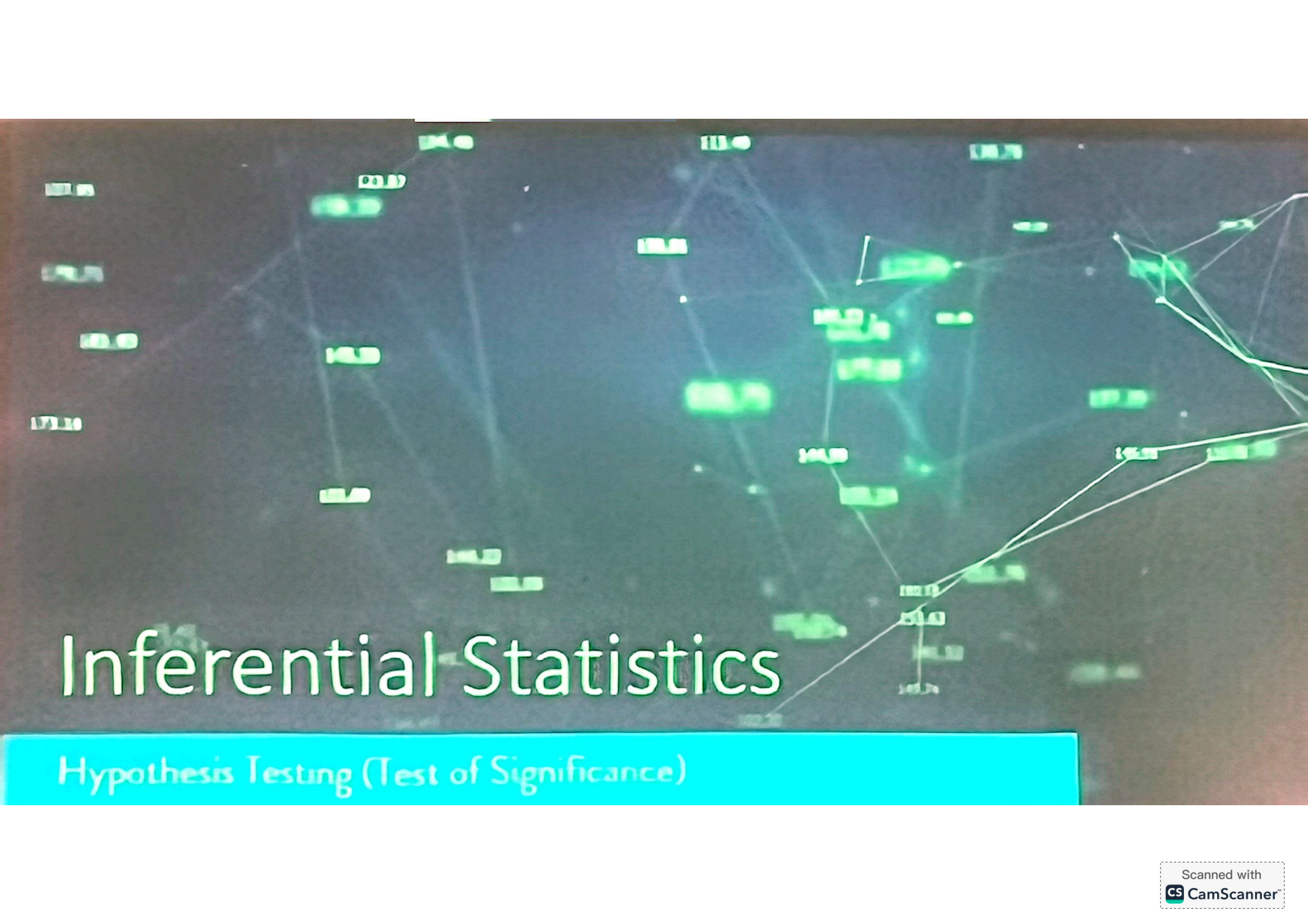


بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

A network graph visualization with numerous green nodes and white connecting lines. The nodes are scattered across the frame, with some appearing as small dots and others as larger, more complex shapes. The lines connect these nodes in a web-like structure, with some lines being straight and others curved. The background is dark and textured.

Inferential Statistics

Hypothesis Testing (Test of Significance)

Hypothesis Testing

- Experiments are designed to test a hypothesis (idea)
- Hypothesis testing uses statistics to determine whether a research hypothesis is true.
- Two possibilities
 - there is a difference due to treatment, or
 - the treatment causes no difference.
- These two possibilities are named the
 - null hypothesis (H_0) and the
 - alternative hypothesis (H_A).

- The null hypothesis (H_0) assumes no change. H_0 is opposite to what a researcher expects.

$$H_0: \mu_1 = \mu_2$$

$$H_0: \mu_1 = \mu_2 = \mu_3 -$$

- The alternative hypothesis (H_A), assumes the existence of a difference. i.e., the H_A is the research hypothesis.

$$H_A: \mu_1 \neq \mu_2$$

$$H_A: \mu_1 \neq \mu_2 \neq \mu_3 -$$

- At the end of the test, the decision will be either:
 - (1) to reject H_0 , or
 - (2) not to reject H_0 .

TRUE OR FALSE

1.

Usually, the H_0 assumes no change and is opposite to what a researcher expects.



2.

The null hypothesis always includes an equal (=) sign.



3.

When a statistical test rejects the null hypothesis (H_0), then the researcher's hypothesis (H_A) becomes not acceptable.



4.

The null hypothesis means that what is being tested has an effect.



Types of Experimental Errors

- We use **samples** in experiments (not the whole populations).
 - Therefore, you cannot be 100% confident that the conclusion from an experiment is the correct one.

Types of Experimental Errors (= Types of Wrong Conclusions)

$$H_0: \mu_1 = \mu_2$$

$$H_A: \mu_1 \neq \mu_2$$

1. Type I error = rejecting H_0 , when H_0 is true (**false +ve**).
2. Type II error = not rejecting H_0 , when H_0 is false (**false -ve**).

- The probability of making a **type I** error is denoted as α .
- The probability of making a **type II** error is denoted as β .
- We can reduce the probability of a **type I error** by making α small (P of type I error = **significance level** = α).
 - The typical values of α are 0.05 (5%) and 0.01 (1%).

Inferential Statistics

Estimation (confidence interval)

- Calculate a population value from observations of a sample drawn from the same population. **Confidence Interval**

Hypothesis testing (tests of significance)

Testing a Single Mean

One sample T test

One Sample (one group) t Test

- To determine whether a sample mean differs significantly from a known (standard) value.
 - Suppose a food safety inspector wants to check if the average nitrite level in a batch of luncheon meat meets the 200 ppm limit."

Assumptions

- The data is
 - normally distributed,
 - the observations are independent,
 - the sample is random.

26.1	19.5	26.0	23.7	21.9	28.0	21.6	25.2	33.6	21.5
33.2	22.1	28.1	20.6	29.4	27.7	22.5	27.5	26.0	24.6

The data represent the average daily milk yield of 20 cows. Is the sample mean significantly different from the population (breed) mean of 26.9 kg?

Solution

(1. Hypotheses,

2. Test,

3. Decision)

1. Hypotheses

$$H_0: \mu = 26.9$$

$$H_A: \mu \neq 26.9$$

2. Test Statistic: t-test (one sample)

- The calculated t value is

$$t = \frac{\bar{X} - \mu}{s/\sqrt{n}}$$

$$(\bar{X} = 25.44, \mu = 26.9, S = 3.92, n = 20)$$

$$t = \frac{\bar{X} - \mu_0}{s/\sqrt{n}} = \frac{25.44 - 26.9}{3.92/\sqrt{20}} = -1.667$$

3. Decision (compare the calculated value to the tabulated value)

General Concept

If the calculated value is $<$ the tabulated (critical) value, H_0 is not rejected
(= the difference is not significant).

If the calculated value is $>$ the tabulated (critical) value, H_0 is rejected
(= the difference is significant).

- At $\alpha = 0.05$ and degrees of freedom $(n - 1) = 19$, the critical (tabulated) value is 2.093.
- Since the calculated $t = 1.667$ is less than the critical value (2.093), H_0 is not rejected. The sample mean is not significantly different from 26.9 kg.

Appendix Table B:

Critical Values of Student T Distribution (two-tailed test)

df / α	0.100	0.050	0.025	0.010	0.005
1	0.914	1.270	1.611	2.151	2.341
2	0.816	1.054	1.385	1.886	2.015
3	0.765	1.000	1.318	1.771	1.895
4	0.741	0.978	1.282	1.734	1.860
5	0.728	0.960	1.250	1.699	1.825
6	0.717	0.945	1.232	1.676	1.802
7	0.708	0.933	1.219	1.658	1.782
8	0.700	0.923	1.209	1.645	1.769
9	0.694	0.915	1.201	1.635	1.759
10	0.689	0.908	1.194	1.627	1.751
11	0.685	0.902	1.188	1.620	1.744
12	0.682	0.897	1.183	1.614	1.738
13	0.679	0.893	1.179	1.609	1.733
14	0.677	0.890	1.176	1.605	1.729
15	0.675	0.888	1.173	1.602	1.726
16	0.674	0.886	1.171	1.600	1.724
17	0.673	0.885	1.169	1.598	1.722
18	0.672	0.884	1.168	1.597	1.721
19	0.671	0.883	1.167	1.596	1.720
20	0.671	0.882	1.166	1.595	1.719
25	0.669	0.880	1.164	1.593	1.717
30	0.668	0.879	1.163	1.592	1.716
40	0.667	0.878	1.162	1.591	1.715
50	0.666	0.877	1.161	1.590	1.714
60	0.666	0.877	1.161	1.590	1.714
70	0.665	0.876	1.160	1.589	1.713
80	0.665	0.876	1.160	1.589	1.713
90	0.665	0.876	1.160	1.589	1.713
100	0.665	0.876	1.160	1.589	1.713
120	0.665	0.876	1.160	1.589	1.713
140	0.665	0.876	1.160	1.589	1.713
160	0.665	0.876	1.160	1.589	1.713
180	0.665	0.876	1.160	1.589	1.713
200	0.665	0.876	1.160	1.589	1.713
250	0.665	0.876	1.160	1.589	1.713
300	0.665	0.876	1.160	1.589	1.713
400	0.665	0.876	1.160	1.589	1.713
500	0.665	0.876	1.160	1.589	1.713
600	0.665	0.876	1.160	1.589	1.713
700	0.665	0.876	1.160	1.589	1.713
800	0.665	0.876	1.160	1.589	1.713
900	0.665	0.876	1.160	1.589	1.713
1000	0.665	0.876	1.160	1.589	1.713

Confidence Interval (CI)

- (CI) is a range of values used to estimate the true value of a population parameter.
- For population mean, the 95% CI is calculated as follows:

$$95\% \text{ CI} = \text{Mean} \pm (t_{0.05} \times \text{SE})$$

2. Test Statistic: t-test (one sample)

- The calculated t value is

$$t = \frac{\bar{X} - \mu}{s/\sqrt{n}}$$

$$(\bar{X} = 25.44, \mu = 26.9, S = 3.92, n = 20)$$

$$t = \frac{\bar{X} - \mu_0}{s/\sqrt{n}} = \frac{25.44 - 26.9}{3.92/\sqrt{20}} = -1.667$$

26.1	19.5	26.0	23.7	21.9	28.0	21.6	25.2	33.6	21.5
33.2	22.1	28.1	20.6	29.4	27.7	22.5	27.5	26.0	24.6

- Example: Milk yields of 20 cows.
- the CI for estimation of the population mean

$$95\% \text{ CI} = \text{Mean} \pm (t_{n, 0.05} \times \text{SE})$$

- Sample mean = 25.44 Sample S = 3.92 n = 20
- At α (p value) = 0.05, and DF = n - 1 (20 - 1 = 19), the critical t value is 2.093
- $\text{SE} = S/\sqrt{n} = 3.92/\sqrt{20} = 0.88$
- $95\% \text{ CI} = 25.44 \pm (2.093 \times 0.88)$
- $95\% \text{ CI} = 25.44 \pm 1.84$ or, 23.6 to 27.3

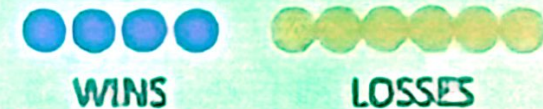
Measuring Association in 2 x 2 Tables

- Quantifying the size of the association
 - Relative risk (RR)
 - Odds ratio (OR).
- Significance of the association (hypothesis testing)
 - Chi-square test.

Odds vs. Probability (Risk)



If a racehorse runs 10 races and wins 4 times.



The probability of the horse winning a race is $4/10 = 0.4$

$$\text{Probability} = \frac{\text{Wins}}{\text{Wins} + \text{Losses}} = \frac{4}{10}$$

The odds of the horse winning a race is $4/6 = 0.67$

$$\text{Odds} = \frac{\text{Wins}}{\text{Losses}} = \frac{4}{6}$$

If a racehorse runs 10 races,

Wins	0	1	2	3	4	5	6	7	8	9	10
Losses	10	9	8	7	6	5	4	3	2	1	0
Trials	10	10	10	10	10	10	10	10	10	10	10
Odds	0	0.11	0.25	0.43	0.67	1.00	1.50	2.33	4.00	9.00	10.0
Probability	0	0.10	0.20	0.30	0.40	0.50	0.60	0.70	0.80	0.90	1

Quantifying Risk

- The chance that an individual who possesses certain traits or is exposed to some factors (risk factors, e.g., smoker) also has a specific disease (condition, e.g., developing lung cancer).
- Risk factors:
 - intrinsic factors e.g., breed, sex, age or weight.
 - extrinsic factors e.g., vaccination, or housing system.

Basic 2 x 2 Table

		Disease or condition (Lung Cancer)		total	
		+ve	-ve		
Risk Factor	Group 1 (Smoker)	a (5)	b (45)	a + b (50)	Risk in smokers $5 / 50 = 0.10$
	Group 2 (Nonsmoker)	c (2)	d (48)	c + d (50)	Risk in nonsmokers $2 / 50 = 0.04$
		a + c (7)	b + d (93)	N (100)	